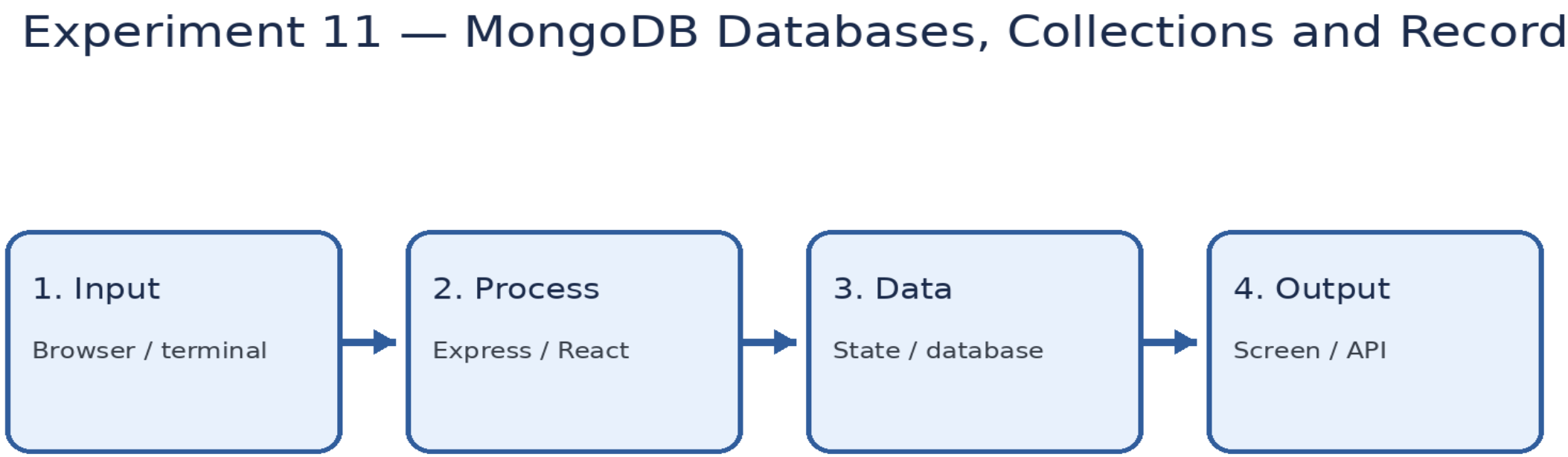
Experiment 11 — MongoDB Databases, Collections and Records

GitHub-readable preview of the presentation. Download the matching .pptx for the editable PowerPoint.

1. Syllabus tasks

- Creating/listing databases and collections.
- `find()`, `findOne()`, filters, projection and count.
- `limit()`, `skip()` and `sort()`.
- Single, compound and unique indexes, index listing and query plan.
- Aggregation with `$match`, `$group`, `$avg`, `$max`, `$push`, `$project`, `$sort`, `$limit`, and `$count`.

2. Workflow



Workflow: input → processing → data/state → output

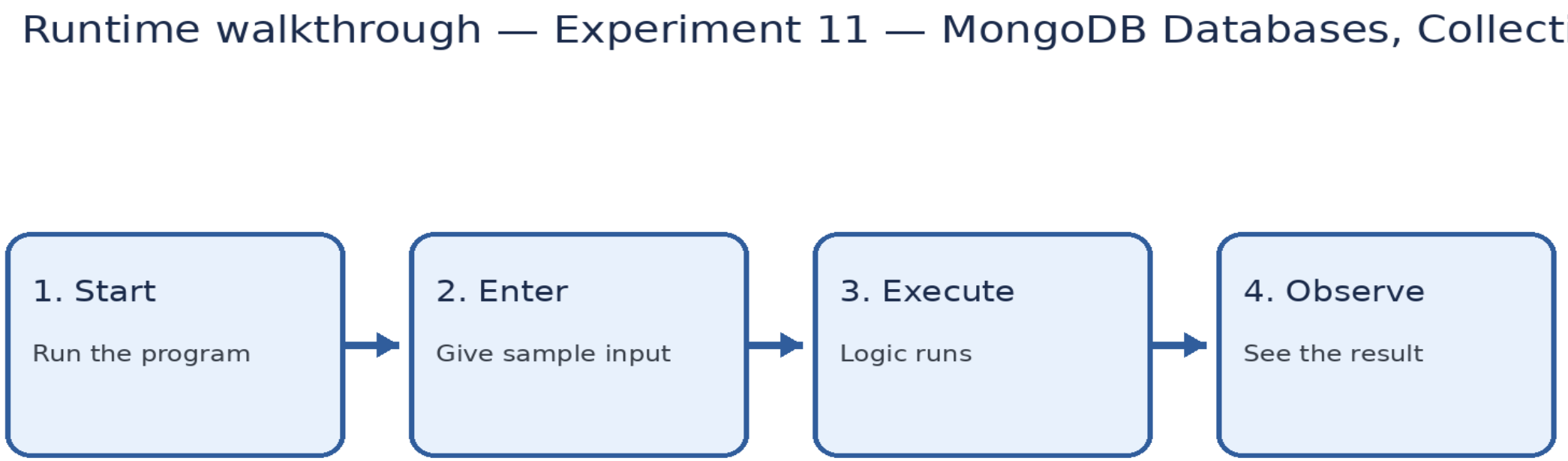
3.1. Code pathway — queries.js

```
/**
 * Experiment 11: MongoDB Databases, Collections and Records.
 *
 * Run from repository root:
 * mongosh --file 11-mongodb-databases-collections-records/queries.js
 * For Atlas:
 * mongosh "YOUR_ATLAS_URI" --file 11-mongodb-databases-collections-records/queries.js
 *
 * Uses the official handout's college database and eight-student dataset.
 */
```

3.2. Code pathway — queries.js

```
11 db = db.getSiblingDB('college');
12
13 print('PART A: database is ' + db.getName());
14 if (!db.getCollectionNames().includes('students')) db.createCollection('students');
15 if (!db.getCollectionNames().includes('courses')) db.createCollection('courses');
16 db.faculty.updateOne(
17 { name: 'Dr. Rao' },
18 { $set: { name: 'Dr. Rao', dept: 'CSE' } },
19 { upsert: true }
20 );
```

4. Runtime walkthrough



Runtime: start → enter → execute → observe

5. Run and observe

- Start the server / Vite app or open the MongoDB shell as described in the experiment README.
- Enter the sample input or call the required route.
- Watch the terminal for logs and errors.
- Verify the browser, API response, or database result.

6. Viva questions

- What is the purpose of this experiment?
- Which input is required?
- Which line or command performs the main operation?
- What output should you observe?
- What happens when the input changes?